

Aviation Baggage Screening

Introduction

We create an airport delay model that, take passengers, airline and baggage screening machine into account, may provide a definitive answer to the question. We approach the modeling problem by beginning with changing discrete passengers into the continuous normal distribution process. Regarding the model is a multi-object problem about screening machine and airline schedule. With schedule revision algorithm to ensure all passengers can be 100% screened with less EDSs and flights delay time. We got airport A should deploy 26 EDSs and airport B should deploy 28 EDSs.

For security reason and the limited fund and space in the airport, modify EDS models to incorporate the use of ETD machines. Aiming for minimum purchase cost, solve the linear programming. We got airport A should deploy 5 EDSs and 98 ETDs and airport B should deploy 5 EDSs and 106 ETDs.

Assumptions

- Each passenger carries two pieces of baggage.
- Ignore airline without peak hour.
- Ignore passengers for the flight without peak hour.
- The order that passengers for the same flight arrive at airport obey normal distribution. Although the accident the passengers appear is random, from the whole accident is under some rules. From Statistics, we can regard the rule is normal distribution.
- the interval between two flights is same
- Once checked baggage was screened, passenger can directly go to the flight without spending extra time on other process.

Table 1 model inputs and symbols

Symbols	Definition
N_{flight}	Number of flights
N_{seat}	Number of seat on each of flight
N_{e_bag}	Bags can be examined by each per hour
E (%)	The time that is operated by EDS
P_{EDS} , P_{ETD}	Price of a EDS machine Price of a ETD machine
$P_{install}$	Price of installing a EDS machine
A_{EDS} (%), A_{ETD} (%)	Accurate in identifying explosive materials on checked bags

Analysis of the Problem

Using the provided data from airport in TIS, we get all airlines within peak hour. With the number of seats on each flight, we get the number of passengers on each flight and all passengers within peak hour. We regard the order of passengers arriving the airport obey normal distribution. A discrete problem can be changed into continuous problem. From it, the number of EDSs can be estimated.

For security's reason, all checked bags must be 100% screened. Screening will consume time, and the airline will be delayed. It seems that the more EDSs, the better results will be. But for the limited number of EDS machines available, we cannot deploy more EDS. So the problem to determine the number of EDSs and airline scheduling can be concluding into a multi-objects problem.

As a multi-objects, it is unreasonable to find the optimize results to satisfy the less EDSs and the timely scheduling. That is to search the feasible results is the aim to establish our model.

Determining the Number of EDSs

Within peak hour the density of passenger arriving and flights departing are all on the climax. We know that if the number of EDSs can work in peak hour, they can guarantee to work out of peak hour.

Using the given statistic results, we know typically that passengers arrive at their flight between 45 minutes and 2 hours prior to their schedule departure time. It last 3 hours, 180 minutes from first passenger arrive at airport to the last flight depart. We assume the start time is 0 minutes. Passengers will arrive at airport from 0 minutes to 135 minutes and flights depart during 120 minutes to 180 minutes. The time-axis was shown in figure 1.

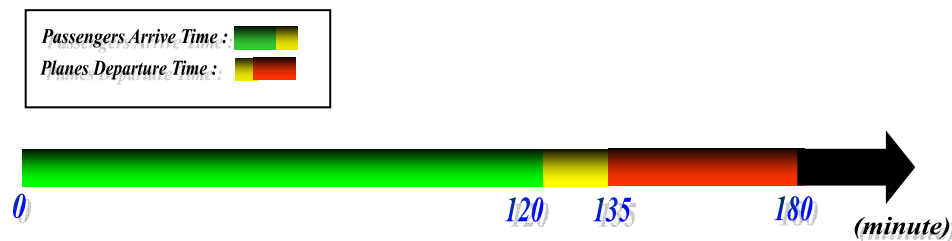


Figure 1 Time-axis

Since a flight cannot depart until all its checked bags are screened using an EDS, to determine the number of EDSs, we must affirm all passengers who prior to the airport can be screened.

From the data in TIS, we got the total number of passengers within peak hour,

$$N_{passenger} = N_{flight} * N_{seat}$$

As the airlines claim that 20% of the passengers don't check any luggage, 20% check one bag, and the remaining passengers check two bags. Using our assumption that each of passengers carries 2 pieces of bag, we got the total amount of checked baggage within peak hour,

$$N_{bag} = N_{passenger} * (0.2 * 0 + 0.2 * 1 + 0.6 * 0.2)$$

For each device is operational about 92% of the time, and between 160 and 210 bags per hour,

considering the worst the device operation, examined bag by each EDS of the operational time.
Examined bag by each EDS of the operational time

$$\alpha = N_{e_bag} * E$$

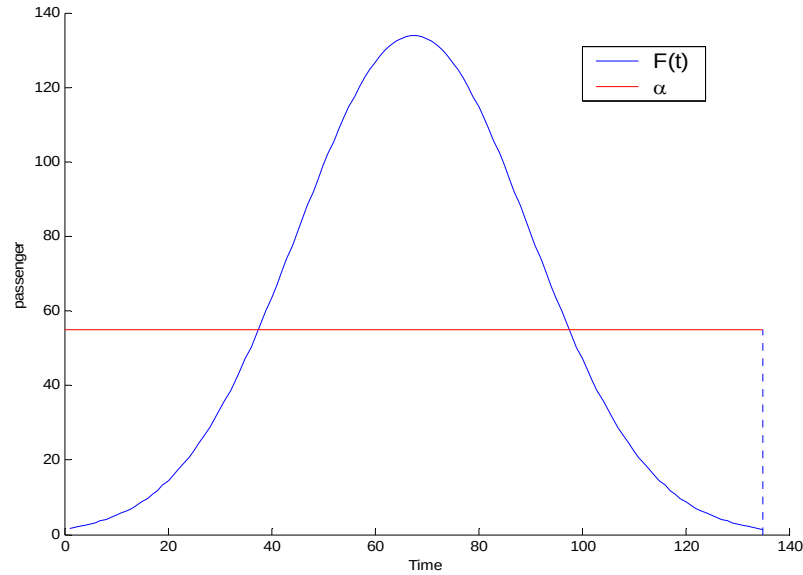


Figure 2

For passengers' arriving obey normal distribution during the 135 minutes, the passengers'

baggage is $F(x) = \frac{N_{bag}}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, from $\int_0^{135} [F(t) - \alpha] dt = 0$ α was gained.

we establish the model to determine the number of EDS,

$$N_{EDS} = \frac{\alpha}{\tilde{\alpha}}$$

Finally, we found the number of EDSs needed in airport A is 23 and in airport B is 25.

Airline Scheduling

Although passengers will arrive prior to the airport, for the security reason, much time are spent on waiting the checked bags to be screened. It results some passengers may delay the flight departure time. For airline has the flexibility to schedule during the peak hour to avoid undesirable flight delays due to unscreened bags.

We can conclude an unreasonable scheduling including two possible circumstances.

- The flight reached the departure time but some passengers were still waiting for screening.
- Some passengers on their seat but their flight were still waiting its departure time.

To avoid the above situations happen, we aim to establish a model to insure the minimum delay time between the flight departure time and its last passenger boarding time.

$$\Delta t_i = T_{fi} - T_{pi}$$

Where, $\{T_{fi}\}$ is current airline schedule, T_{pi} is the last passenger boarding time of No i flight.

T_{pi} is composed with the arriving airport time $\hat{T}_{pi}$ and the time waiting to screen T_w for the last passenger.

$$T_{pi} = \hat{T}_{pi} + T_w$$

Where, $\hat{T}_{pi} = T_{fi} - 45$,

$$T_w = \frac{g(\hat{T}_{pi})}{\tilde{\alpha}}$$

$g(x)$ is the average queuing passengers before the just arriving passenger, as it was shown in figure 3

$$g(x) = \begin{cases} \frac{1}{N_{EDS}} \int_{x_0}^x (F(x) - \alpha) dt & (x_0 < x \leq 135) \\ 0 & (0 \leq x \leq x_0) \end{cases}$$

$F(x)$ is the number of passengers arriving at the airport at a certain time,

$$F(x) = \sum_{i=1}^k f_i(x)$$

Where, $f_i(x)$ is the number of passengers for the same flight,

$$f_i(x) = \frac{N_{passenger-i}}{\sqrt{2\pi a} * \sigma_i} e^{-\frac{(x-u_i)^2}{2\sigma_i^2}}$$

$$while \quad u_i = \frac{a_i + b_i}{2},$$

$$\sigma_i = \frac{b_i - a_i}{6}$$

$$a_i = T_{fi} - 120,$$

$$b_i = T_{fi} - 45$$

and solving the equation $F(x) = \alpha$, we can gain x_0

We establish airport delay Model

$$MIN \quad |\Delta t_i|$$

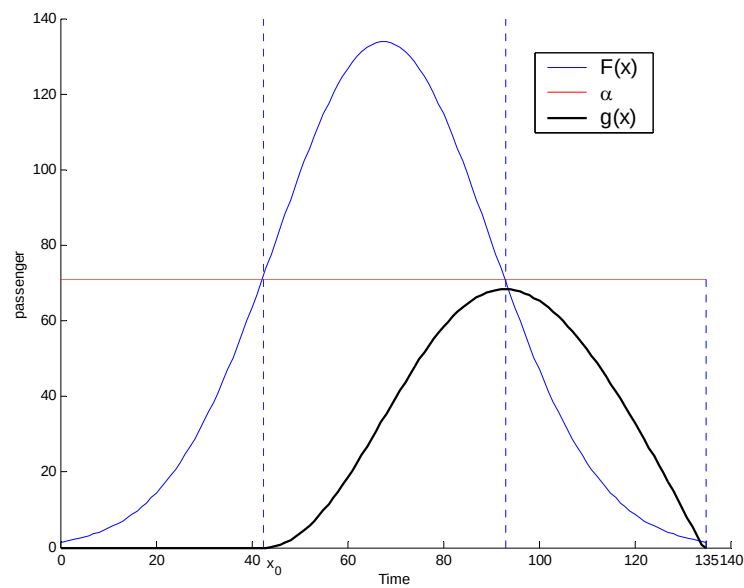


Figure 3 passengers arriving at airport ($F(t)$) and passengers waiting behind per EDS ($g(t)$) at t time

Table 2

N_{EDS}	Number of Delay flights
23	11
24	6
25	2
26	0

Table 3

N_{EDS}	Number of Delay flights
25	9
26	5
27	1
28	0

From Table 2 and Table 3, we could see no flights delayed when there were 26 EDSs in airport A and 28 EDSs in airport while there were 11 flights delayed in airport A and 9 in airport B.

Table 4 Flight Departures Schedule for Airport A

Type	Depart (minute)	Queue position	Delta	Type	Depart (minute)	Queue position	Delta
1	172	41	45.0000	5	148	22	29.8875
1	135	12	16.8875	5	176	44	45.0000
1	163	34	44.8875	5	161	32	42.8875
1	124	4	5.8875	5	157	29	38.8875
1	141	17	22.8875	5	132	10	13.8875
1	175	43	45.0000	5	126	5	7.8875
1	162	33	43.8875	5	142	18	23.8875
1	150	24	31.8875	5	145	20	26.8875
1	133	11	14.8875	5	128	7	9.8875
1	170	39	45.0000	5	146	21	27.8875
2	140	16	21.8875	5	137	14	18.8875
2	165	35	45.0000	5	136	13	17.8875
2	171	40	45.0000	5	139	15	20.8875
2	159	31	40.8875	5	174	42	45.0000
3	152	25	33.8875	5	169	38	45.0000
3	123	3	4.8875	5	153	26	34.8875
3	166	36	45.0000	6	129	8	10.8875
4	127	6	8.8875	6	179	46	45.0000
4	120	1	1.8875	6	158	30	39.8875
4	167	37	45.0000	6	144	19	25.8875
5	156	28	37.8875	6	149	23	30.8875
5	122	2	3.8875	7	154	27	35.8875
5	131	9	12.8875	8	178	45	45.0000

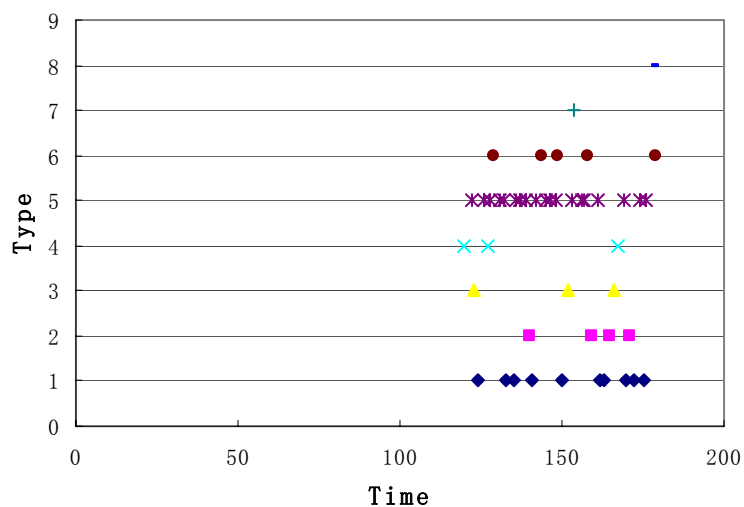
**Figure 4 Flights of airport A depart during peak hour**

Table 5 Flight Departures Schedule for Airport B

Type	Depart (minute)	Queue position	Delta	Type	Depart (minute)	Queue position	Delta
1	150	25	32.5075	4	123	3	5.5075
1	142	18	24.5075	4	133	11	15.5075
1	170	41	45.0000	5	145	21	27.5075
1	128	7	10.5075	5	177	46	45.0000
1	154	28	36.5075	5	167	38	45.0000
1	149	24	31.5075	5	130	9	12.5075
1	162	34	44.5075	5	127	6	9.5075
1	147	22	29.5075	5	122	2	4.5075
2	143	19	25.5075	5	163	35	45.0000
2	132	10	14.5075	5	140	17	22.5075
2	148	23	30.5075	5	173	43	45.0000
2	178	47	45.0000	6	165	37	45.0000
2	144	20	26.5075	6	152	26	34.5075
2	153	27	35.5075	6	169	40	45.0000
3	125	5	7.5075	6	124	4	6.5075
3	160	33	42.5075	6	129	8	11.5075
3	157	30	39.5075	6	134	12	16.5075
3	135	13	17.5075	6	174	44	45.0000
3	120	1	2.5075	6	155	29	37.5075
3	168	39	45.0000	6	158	31	40.5075
3	139	16	21.5075	6	164	36	45.0000
4	172	42	45.0000	7	138	15	20.5075
4	137	14	19.5075	7	159	32	41.5075
4	175	45	45.0000	8	179	48	45.0000

Advices

- **To Mr. Sheldon**

As data in table 2 and table 3 show that increase the number of EDSs would decrease the delayed flight. Considering the costs and space a EDS need, fewer delayed flights with 25 EDSs in airport A and 27 EDSs in airport B would be more suitable.

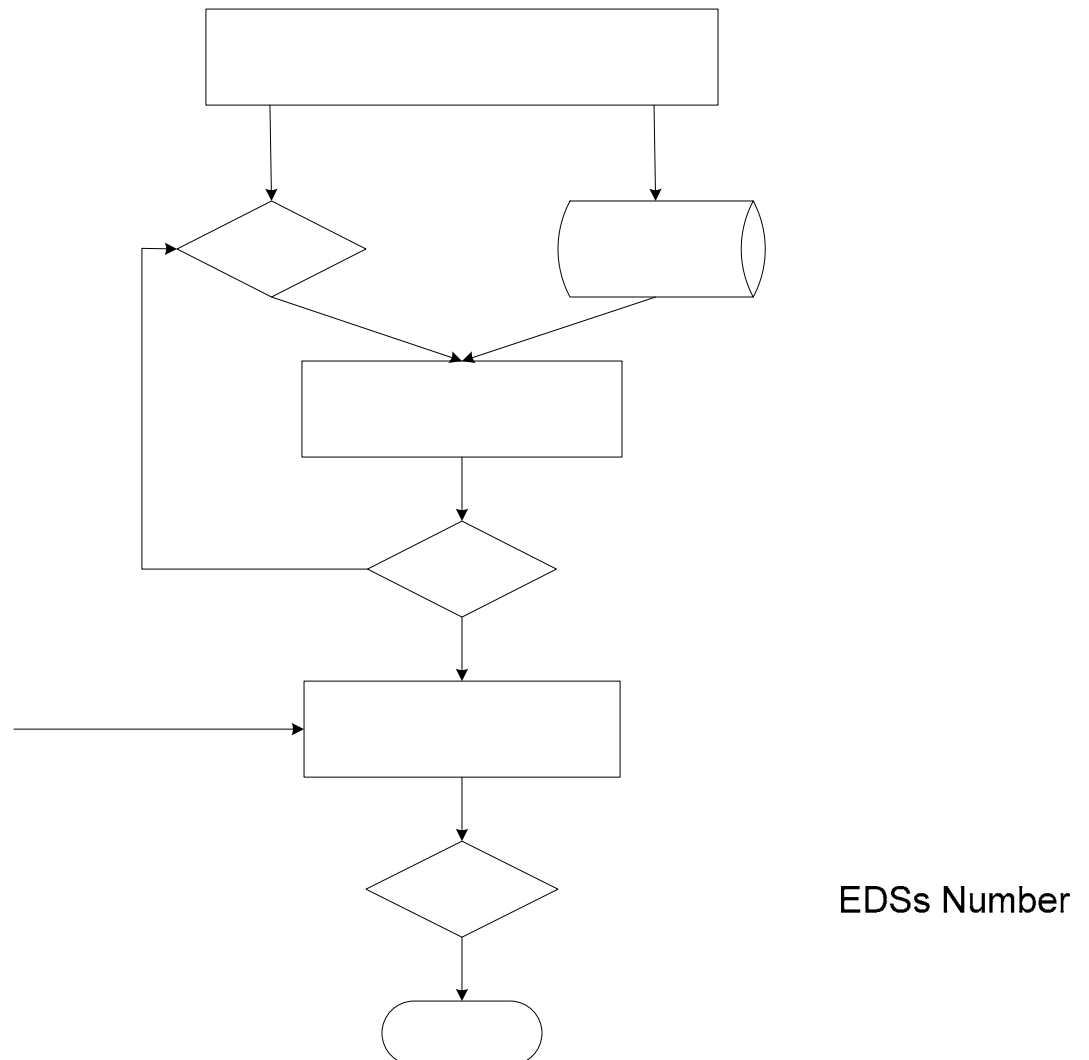
- **For airport**

Ask passenger arrive ahead of schedule more early will also decrease the delayed flight.

Memo to Office of Security Operations

Good results of the number of EDSs and airline scheduling generated by our model. The chart flow 1 indicate the process that how the model works. First, get the data from the TIS. The data are simple and no special and can be easily got. Under our assumption, the order that passenger arriving airport obey distribution, etc. We estimate the Number of EDSs number. With the number

and current schedule, which can be generated randomly or transplanted from a real system. To aim for the least delayed time of a flight. An airport delay model is built. Then we got the schedule for flight delay. The EDS has a strong effect on flight delay, which also can affect the EDS. With schedule revision algorithm, we can adjust the airline schedule. So future schedule will be a better one than ever.



Flow chart 1

The model has strong transplant and flexibility. Our model is presumably close to the actual profile.

Displacement EDS with ETD

Adding ETD machines into the system, there were 3 ways, In-line EDS, Pure ETD and Mix EDS and ETD machines. We modify the EDS model and establish the ETD models and expose the results in table 6 and table 7.

- **In-Line EDS**

To meet the require that up to 20% of the passenger need have their checked bags screen

through both an EDS and an ETD machine. We employ m ETDs with the EDSs in airport.

$$m = \frac{N_{bag} \cdot 20\%}{135\tilde{\alpha}}$$

- **Pure ETD**

Each ETD machine costs \$45,000 to purchase comparing with each EDS machine costs \$ 1 million to purchase. Try to displace all EDS with ETD.

$$N_{ETD} = m + \frac{\tilde{\alpha}N_{EDS}}{\tilde{\alpha}_2}$$

- **Mixed ETD and EDS Machines**

The interim scheme is to mix ETD and EDS in the system. We build a model to minimum the costs of EDS and ETD to purchase.

$$\begin{aligned} \min P &= P_{EDS} \tilde{N}_{EDS} + P_{ETD} \tilde{N}_{ETD} \\ S.T. \quad &\begin{cases} \tilde{N}_{EDS} \cdot \tilde{\alpha} + \tilde{N}_{ETD} \cdot \tilde{\alpha}_2 = \alpha + \frac{N_{bag} \cdot 20\%}{135} \\ \tilde{N}_{EDS} \geq \frac{N_{bag} \cdot 20\%}{135\tilde{\alpha}} \\ \tilde{N}_{ETD} \geq \frac{N_{bag} \cdot 20\%}{135\tilde{\alpha}_2} \end{cases} \end{aligned}$$

Table 6 Different ETD proportion in airport A

Item	In-Line EDS	Pure ETD	Mixed ETD and EDS Machines
Equipment	26 EDS	0 EDS	5 EDS
	18 ETD	116 ETD	98 ETD
Purchase Costs	\$26,810,000	\$5,220,000	\$9,410,000

Table 7 Different ETD proportion in airport B

Item	In-Line EDS	Pure ETD	Mixed Trace and EDS Machines
Equipment	28 EDS	0 EDS	5 EDS
	19 EDT	125 ETD	106 ETD
Purchase Costs	\$28,855,000	\$5,625,000	\$9,770,000

As the results showed in table above, we can see the purchase costs on mixed ETD and EDS machines only 35% of the purchase costs on In-line EDS in airport A and 33% in airport B. So it is worth to replace the ETDs with part of the EDS devices. In addition, it is quite cheap to employ pure ETD.

To analysis the mix proportion of EDS and ETD, we bring in a new evaluation modulus:

$$\xi_{EDS} = \frac{1,000,000 + N_{year} * P_{EDS_operate}}{\alpha}$$

$$\xi_{ETD} = \frac{45,000 + N_{year} * P_{ETD_operate}}{\alpha_2}$$

where $10 * P_{EDS_operate} = P_{ETD_operate}$

- ξ_{EDS}, ξ_{ETD} are respectively the average cost of each EDS,ETD screening a bag
- α, α_2 are respectively screening efficiency of EDS,ETD
- $P_{EDS_operate}, P_{ETD_operate}$ are respectively operation expenses of EDS,ETD operation expenses each year

When $\xi_{EDS} = \xi_{ETD}$, Solution is $N_{year} * P_{EDS_operate} = 21026$

So we can learn that N_{year} is between 2.3362 and 10.5128 year, when $P_{EDS_operate}$ is between 2000 and 9000. That is to say, when the using year is smaller then N_{year} , purchasing more ETD is apparently a good idea. And when longer then N_{year} , we should consider deploy A quantitative EDS to decrease the cost

Effects on EDS/ETD Model

As the time goes by, the device technology will be improved, new device with better efficiency cannot but take the place of the old device. And we must realize that, if we do not wash out the old devices, we will cost a lot of money to stick up for the old device with lower efficiency. If we use the money that spend in the maintenance for the old device to buy new devices with better efficiency and lower price, we will benefit from it and get the same or event better efficiency. But the change of the device technology will not impact the principle of our model, because the progress of the device technology no more than the changes of the efficiency and the cost of the device. For this model, these changes are only some modification of parameters' value which has nothing to do with the principle of this model. It is same as the change of accuracy, speed, and operational reliability. They are only parameters in this model.

But for some types of devices that need operators, we must analyze it specially. Because not only the device need the cost of maintenance, but also the operators need labor and capital. For example, the ETD is such a kind of device. Although the cost of this device is cheap compared with EDS, but as the time goes by the cost of labor and capital of the ETDs will be more expensive than the total cost of the EDSs. If this circo occurs, we need think of abandoning the ETDs.

Strengths

- This model adequately thinks of the actual airport's status in quo and all kinds of exigencies, so it has better adaptability.
- With some modification, it can be applied in an actual environment to work out the questions about the purchase strategies of the explosive detection systems (EDS) and explosive trace detection (ETD) machine for an airport.

- At the same time, this model can supply a perfect schedule of the departure of different types of flights according to the number of EDS and ETD. If the director of the airport does not content with the schedule set by this model or something unexpected impact the flights, this model also can supply an adjusting method.
- In our paper a system to build the schedule is given. Only some common parameters needs with this method. Through initial the schedule, with evaluation system and revision algorithm, we can output a satisfied schedule. The system can translate into many kinds of airport. The method is simple, the function is strong.
- The operation of this model is so easy that the operator need not to know the idiographic structure of the arithmetic. All that he needs to do is only the entering of some parameters' value.

Weaknesses

- In order to simplify the model, we ignore the passengers whose flight is out of the peak hour, which may impact the exactness of our model.
- Because of our arithmetic, during the program executing, it may cost a lot of time .
- The model can not supply the absolute perfect solution , but it can supply a comparatively best solution.

References

- [1]. Bags per personal.
<http://api.hq.faa.gov/Conference/Conference2002commercial/Steinert/sld016.htm>. Accessed February 7, 2003
- [2]. Normal distribution. Jinlin Shou. 2002. Mathematic Modeling. 2nd. Xi'an: Xi'an Jiaotong University Press
- [3]. Progress in Implementing Provisions of the Aviation and Transportation Security Act. Statement of Alexis M. Stefani Assistant Inspector General for Auditing U.S. Department of Transportation